

# Beyond RNA and DNA levels and modifications

NeuroGenomics class 13

If all you have is a hammer,  
everything looks like a nail

~ Law of the instrument



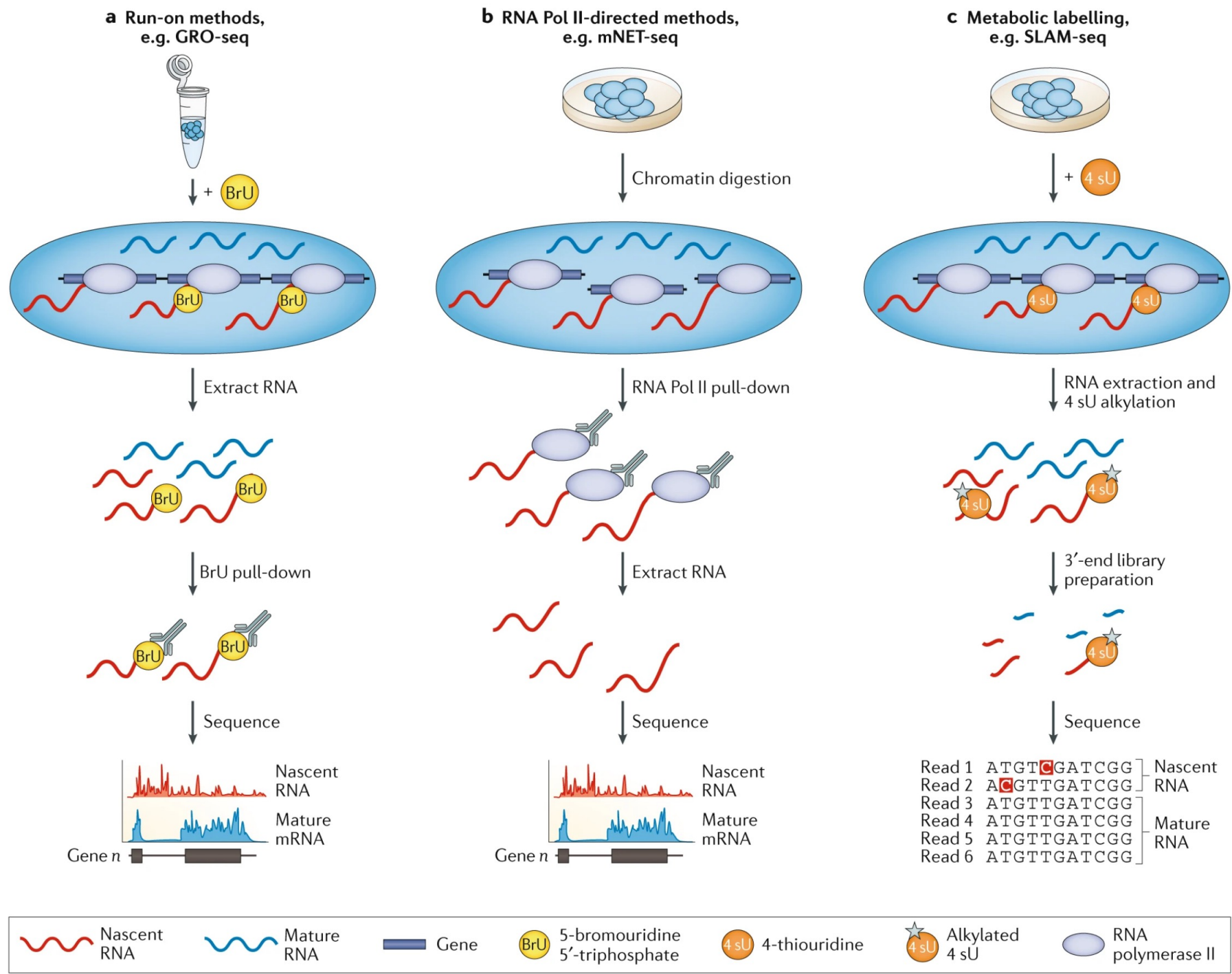
# Outline

- Studying RNA dynamics – sequencing nascent RNA
- Studying RNA dynamics – RNA velocity
- Studying RNA secondary structure
- Studying protein synthesis – ribosome profiling
- Detect open chromatin regions - ATAC-seq
- Studying DNA 3D structure – Hi-C
- Studying neuronal connectivity and activity

# Beyond steady-state RNA analysis

- Standard RNA sequencing reveals a snapshot of stable, or steady-state, RNA levels
- The steady state is the balance of RNA transcription into nascent RNA, RNA processing into mature RNA, and finally RNA degradation
- If we want to learn about genes that are turned on in a dynamic way, we need to measure or estimate the nascent RNA

# Measuring nascent RNA

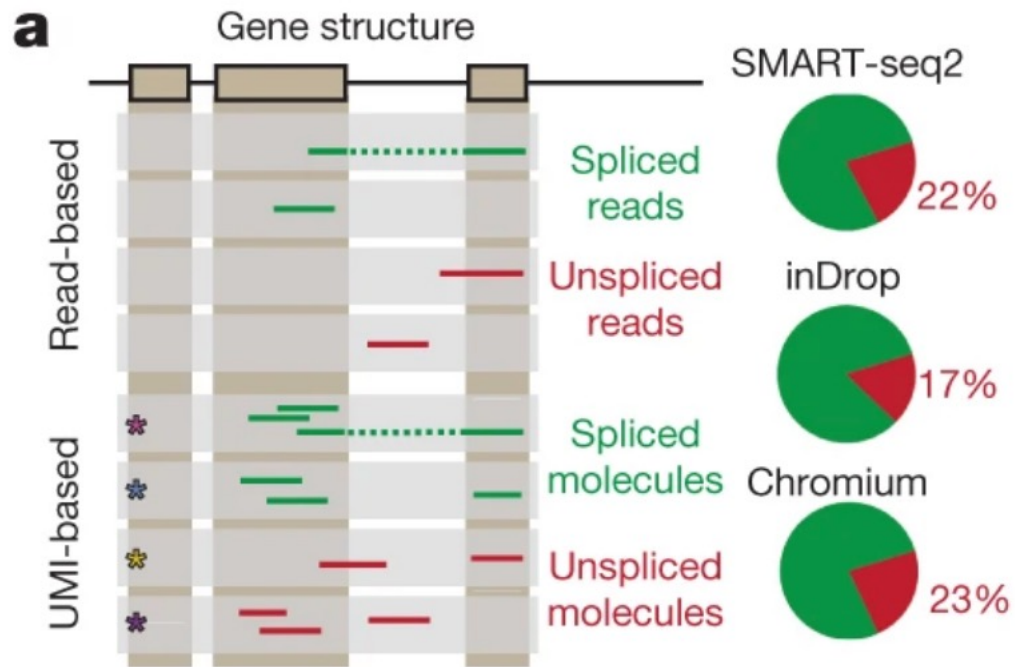


Stark et al., 2019,  
Nature Review Genetics

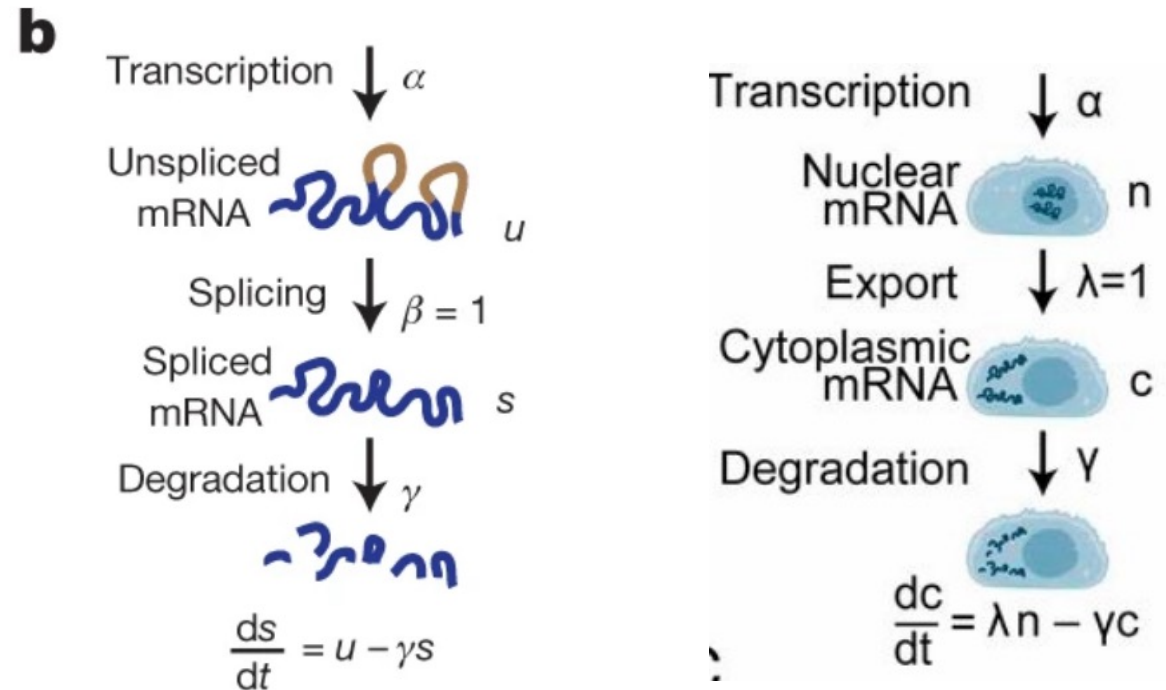
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# The concept of RNA velocity – modelling RNA creation/degradation to predict future cell states



Manno et al., 2018, Nature



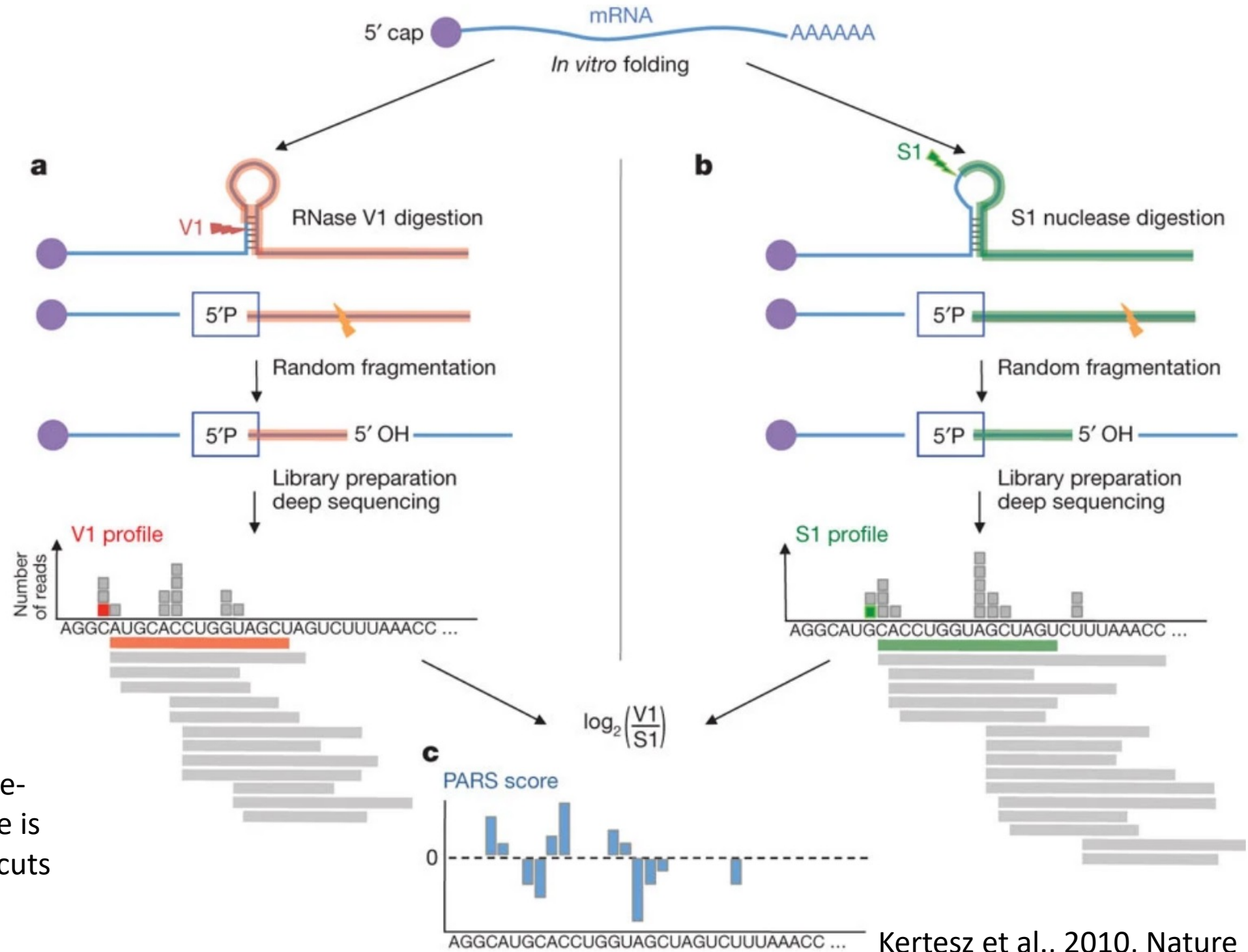
Xia et al., 2019, PNAS

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# Studying RNA secondary structure



a) RNA molecules are cleaved by RNase V1, which cuts 3' of double-stranded RNA. B) the RNA sample is treated with S1 nuclease, which cuts 3' of single-stranded RNA.

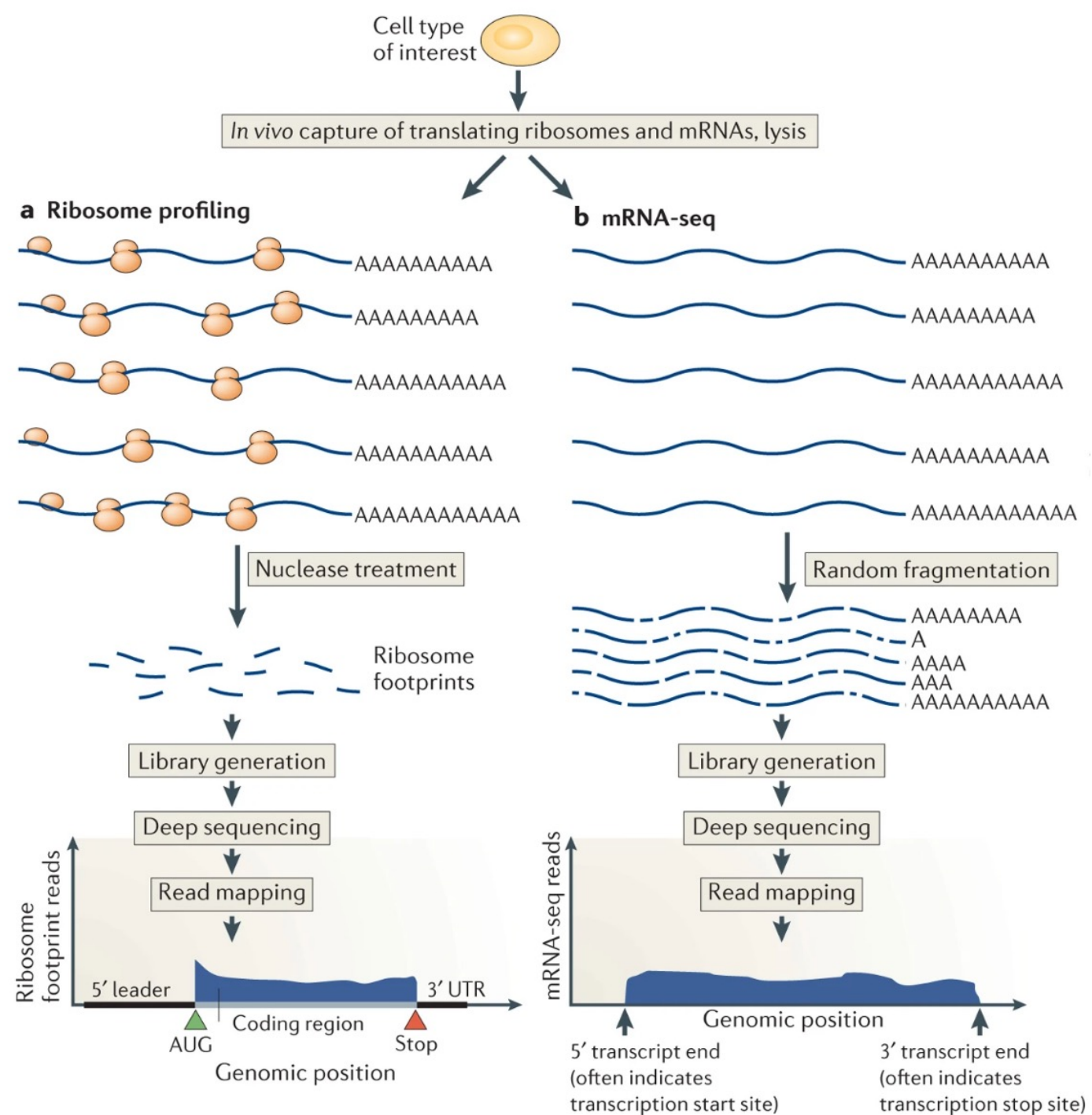
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# Studying protein synthesis with ribosome profiling

Ribosome-bound mRNAs are isolated by size and treated with RNase, resulting in protected mRNA fragments termed 'footprints'

Brar and Weissman, 2015  
Nature Reviews  
Molecular Cell Biology



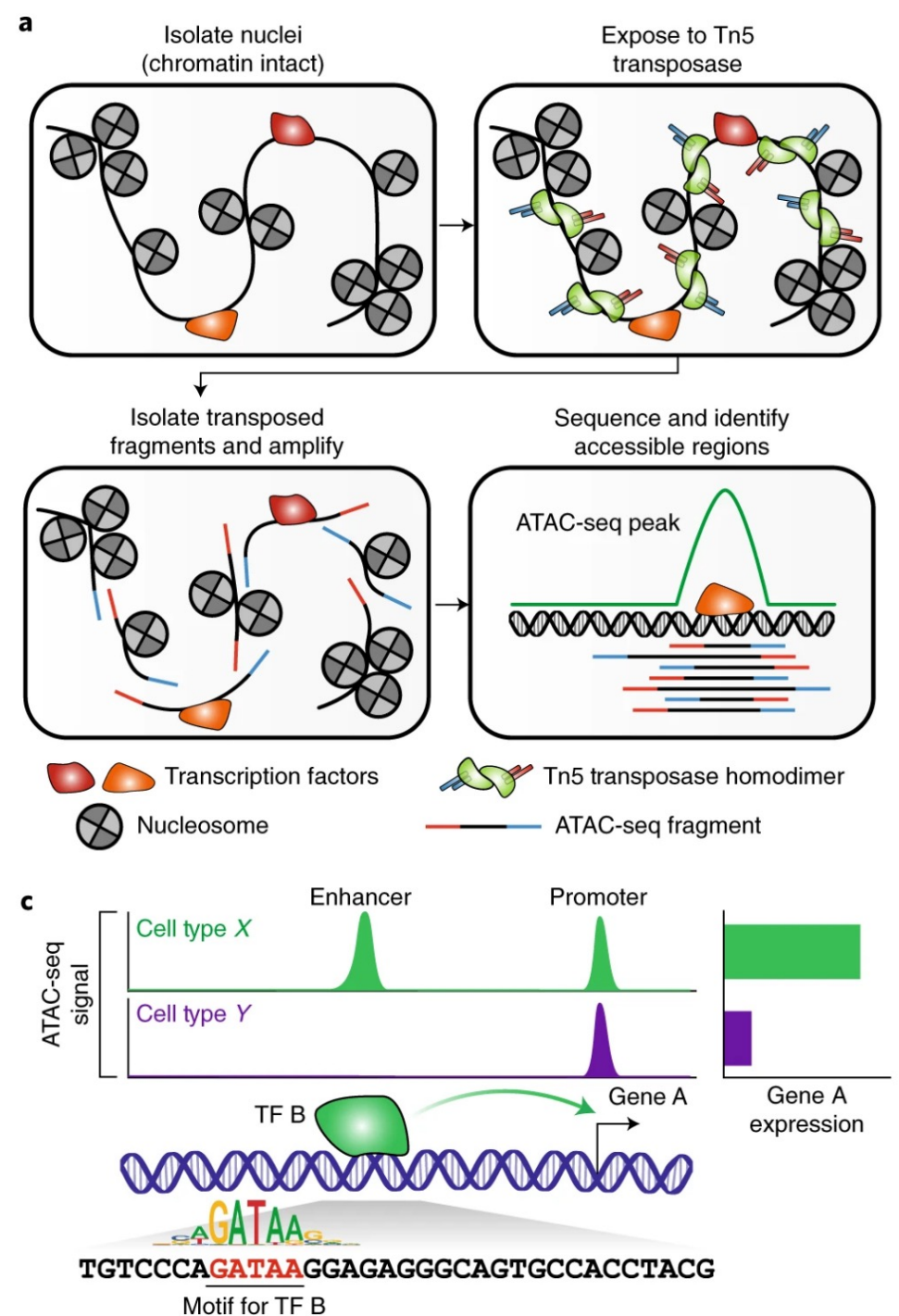
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# Detect open chromatin with ATAC-seq

- Nuclei are isolated from cells, maintaining the chromatin structure and any associated DNA-binding proteins, including nucleosomes and TFs, intact. This chromatin is then exposed to the Tn5 transposase, which acts as a homodimer to simultaneously fragment the chromatin and insert sequences containing PCR handles
- ATAC-seq in situ?

Grandi et al., 2022, Nature Protocols



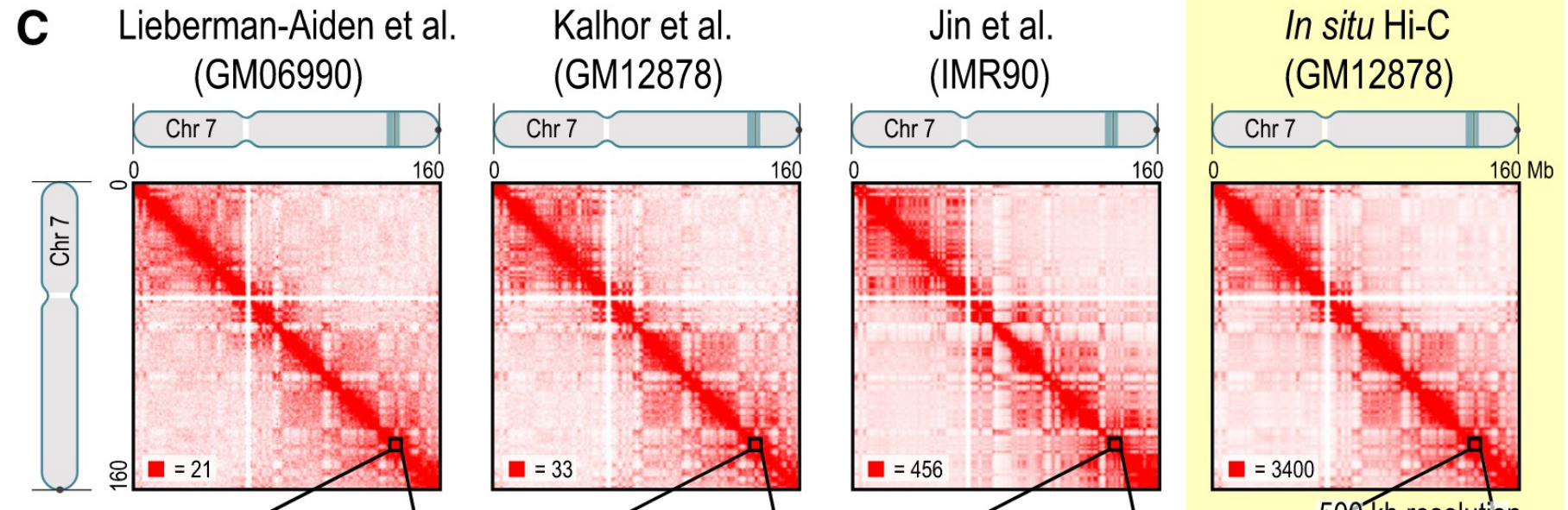
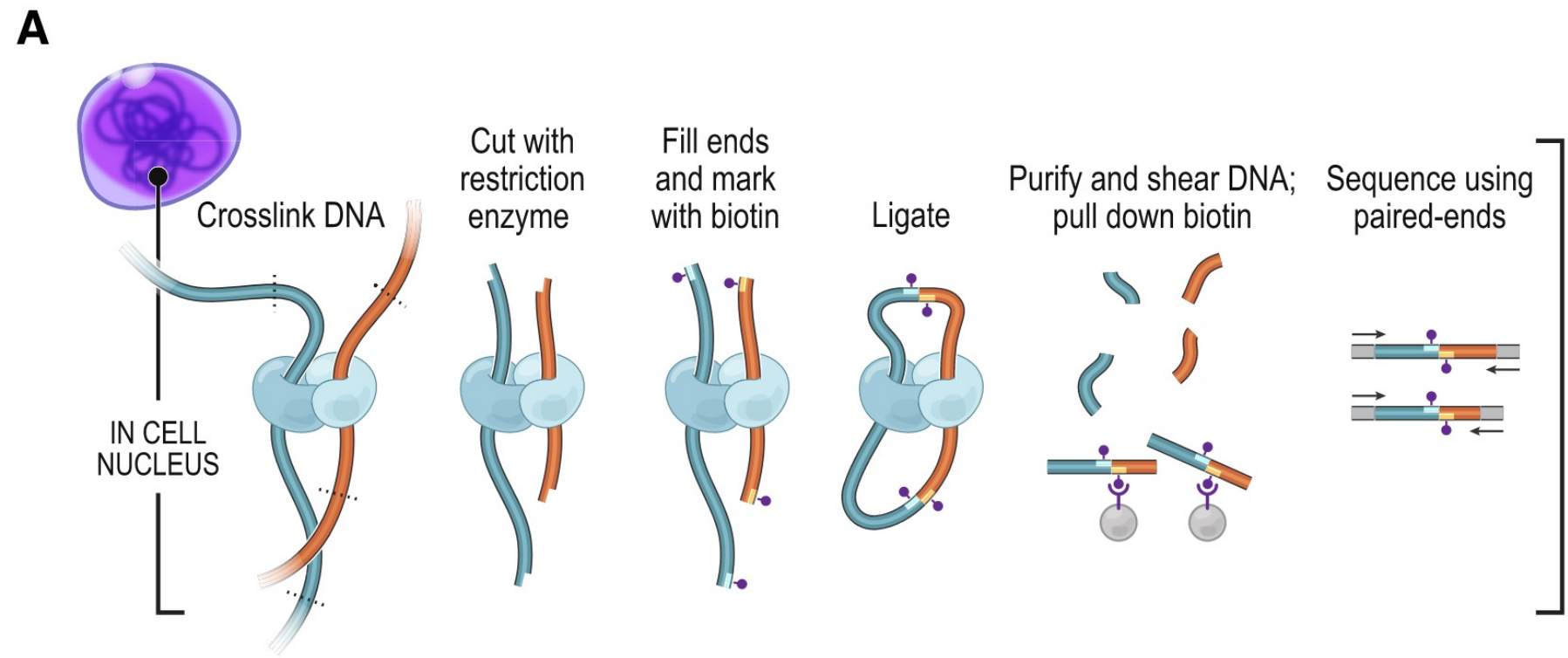
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# Studying the 3D structure of the DNA

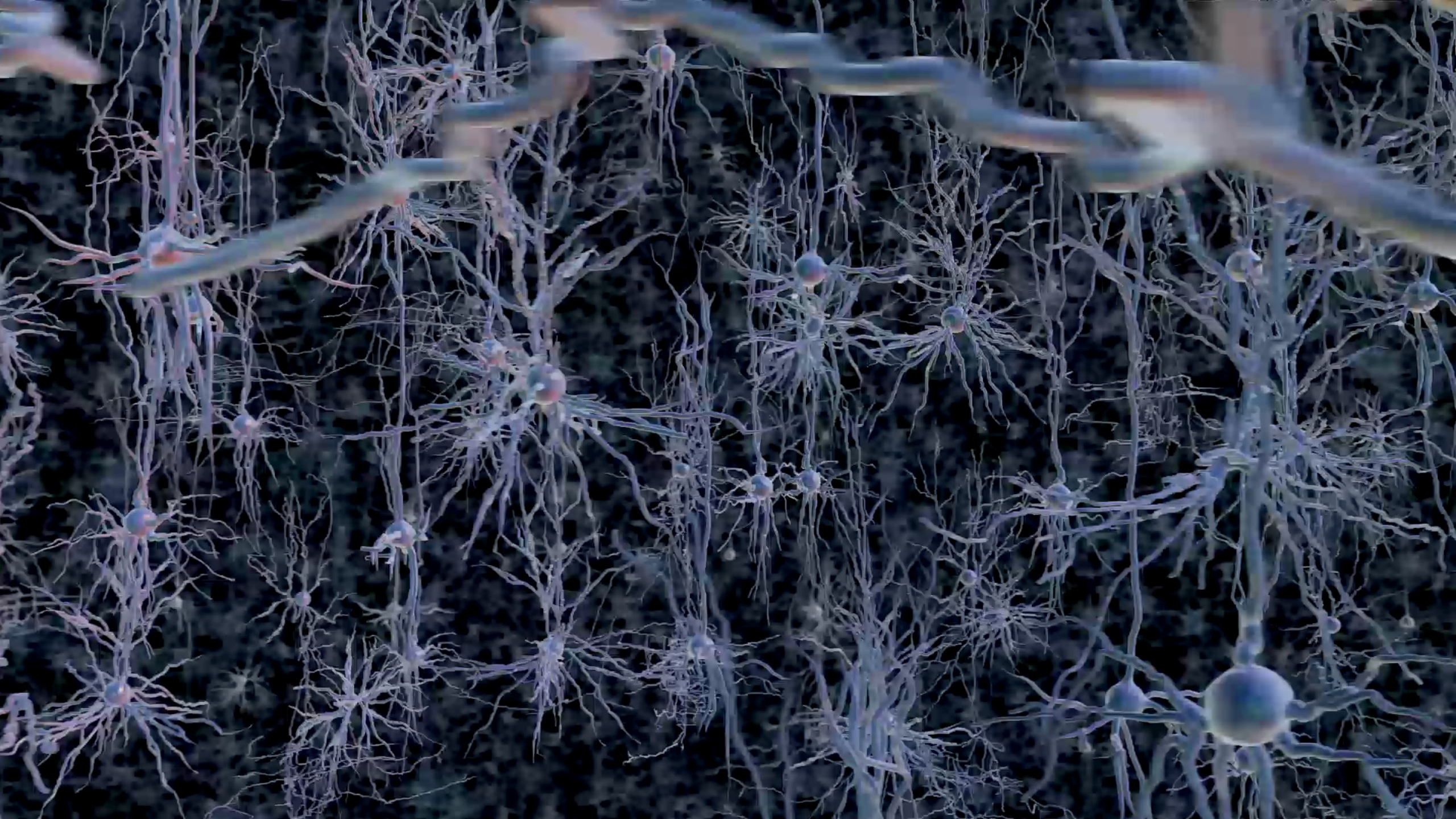
The 3D conformation of chromosomes is bringing widely separated functional elements into close spatial proximity  
Rao et al., 2014, Cell



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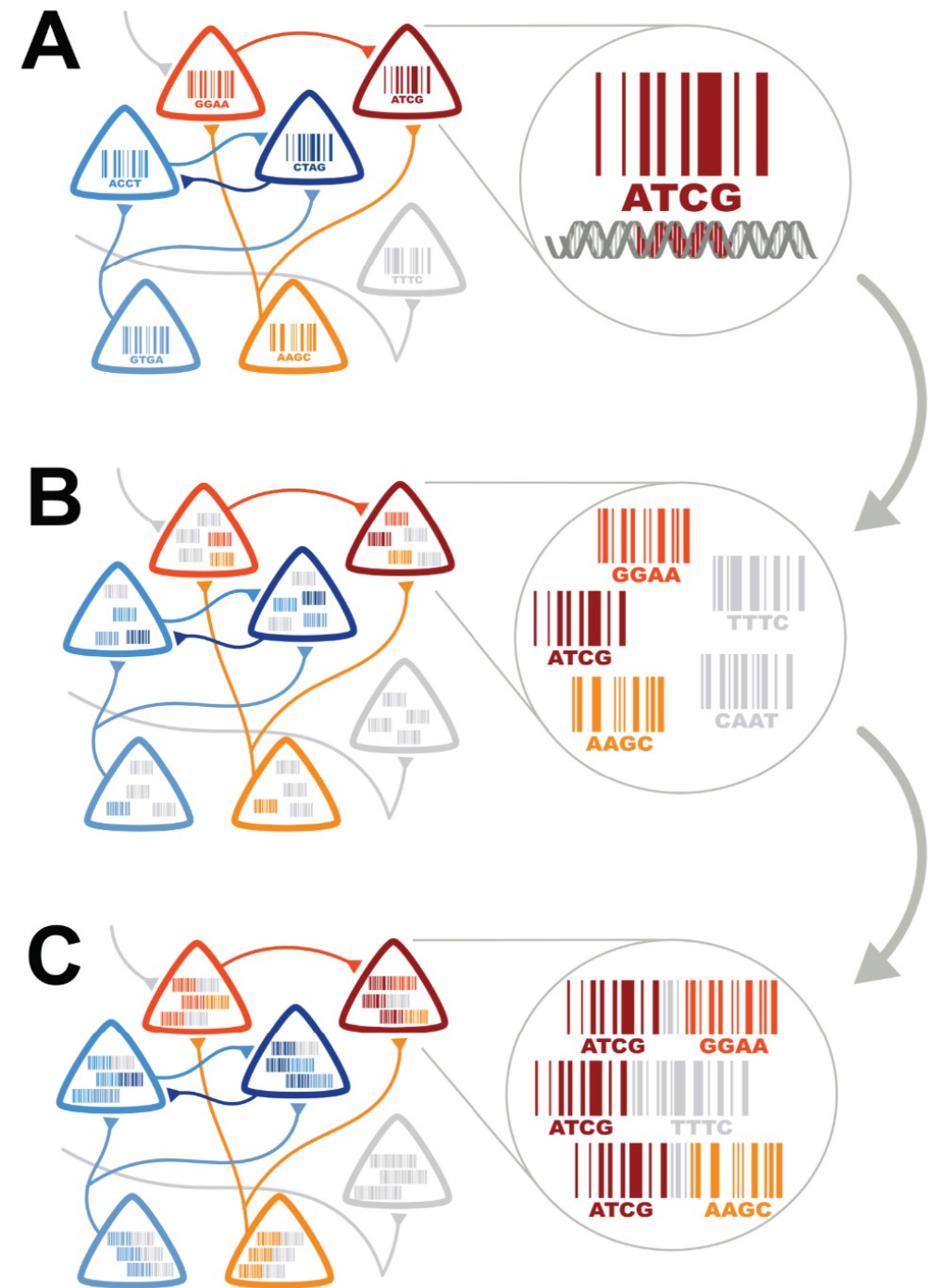




# Mapping the connectome with sequencing

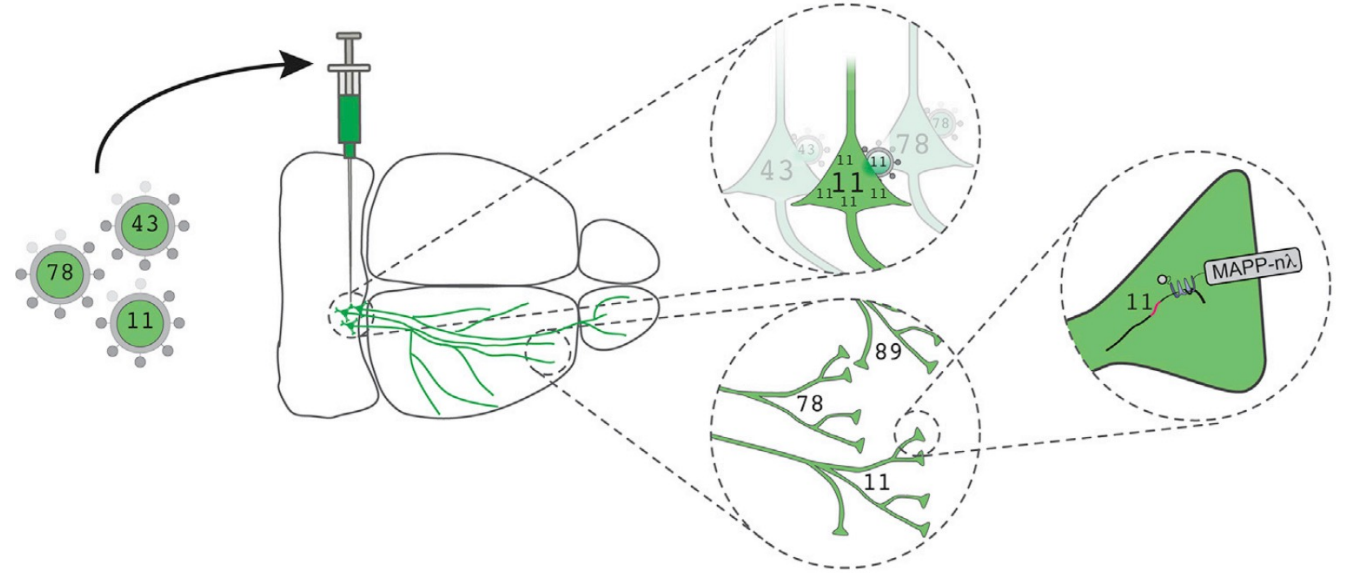
(A) First, each neuron must be labeled with a unique sequence of nucleotides—a DNA “barcode”. (B) Second, barcodes from synaptically connected neurons must be associated with one another, so that each neuron can be thought of as a “bag of barcodes”: copies of its own “host” barcode and copies of “invader” barcodes from synaptic partners. (C) Finally, host and invader barcodes must be joined into barcode pairs. These pairs can be subjected to high-throughput sequencing.

Zador et al., 2012, PLOS biology

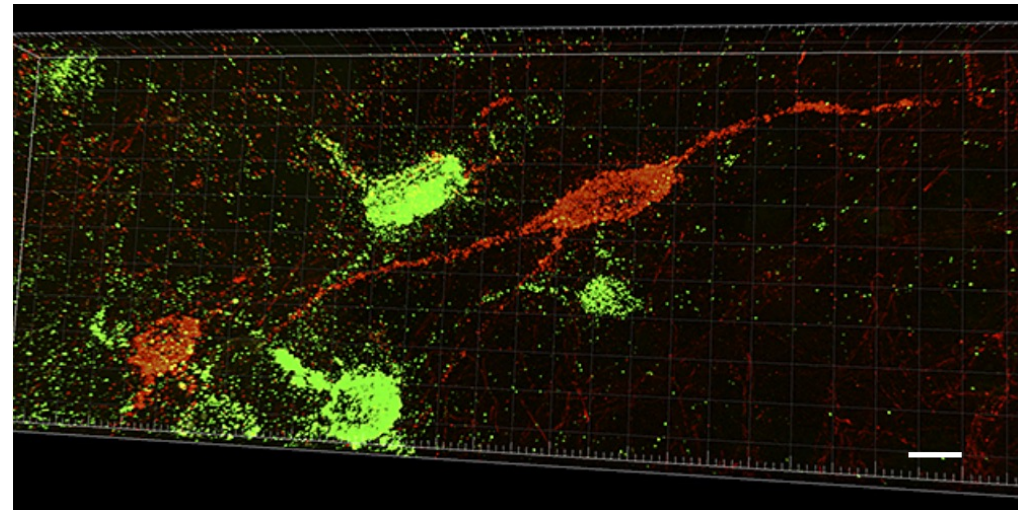
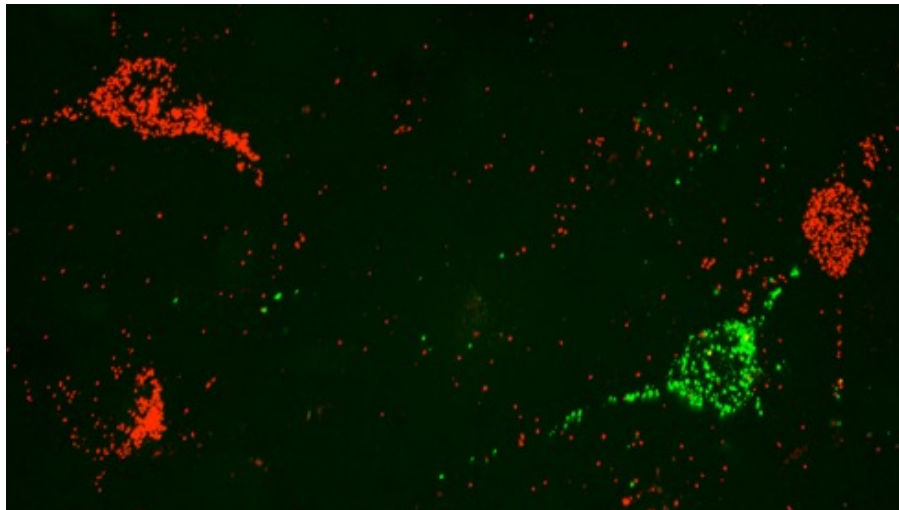


# Mapping the connectome with sequencing

MAPseq = Kebschull et al., 2016, Cell



Collaboration with the Zador lab at Cold Spring Harbor, and the Church lab at Harvard



# Measuring the activity of all the neurons with sequencing?

## Measuring Cation Dependent DNA Polymerase Fidelity Landscapes by Deep Sequencing

Bradley Michael Zamft , Adam H. Marblestone , Konrad Kording, Daniel Schmidt, Daniel Martin-Alarcon, Keith Tyo, Edward S. Boyden, George Church 

